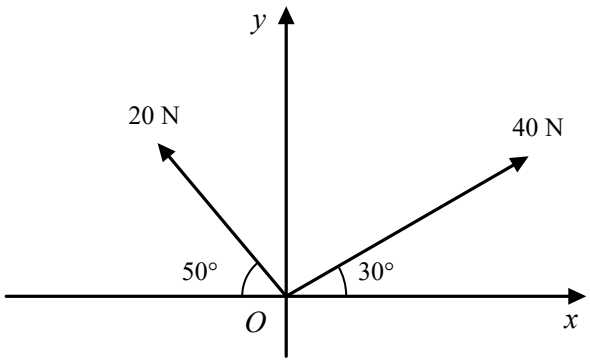
STPM 2021 PHYSICS 960/1

Questions and Answers

All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper.

Section A [15 marks]

1. Two forces, 20 N and 40 N, which act on a point, *O*, are shown in the diagram below.



What are the magnitude and direction of the resultant force with respect to positive *x*-axis?

	Magnitude/N	*Direction/°*
A	41	32
B	41	58
C	59	32
D	59	58

Answer: B

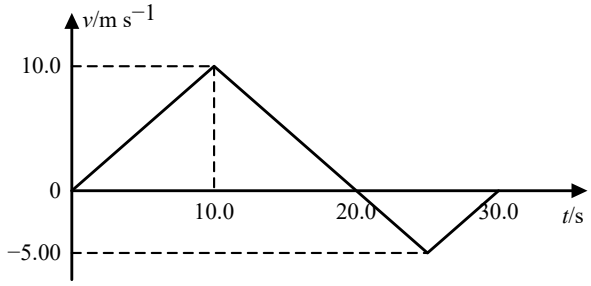
$$F_x = 40\cos 30° - 20\cos 50° = 34.6 - 12.9 = 21.8\ \text{N}$$

$$F_y = 40\sin 30° + 20\sin 50° = 20.0 + 15.3 = 35.3\ \text{N}$$

$$F = \sqrt{(21.8^2 + 35.3^2)} = 41\ \text{N};\ \theta = \tan^{-1}\frac{35.3}{21.8} = 58°$$

Explanation: Resolve each force into *x* and *y* components, taking care with signs: the 20 N force points to the left, so its *x* component is negative. Then combine the components. Forgetting the negative sign gives 59 N (options C and D); the direction is measured from the positive *x*-axis, so it is 58°, not 32°.

2. The variation of velocity, v, with time, t, of a cyclist who travels along a straight road is shown in the graph below.



What is the distance travelled by the cyclist from the starting position to the position that he finally stops?

A 25.0 m B 75.0 m C 100 m D 125 m

Answer: B

Area under the v–t graph: 0 to 20 s: $\frac{1}{2}(20.0)(10.0) = 100$ m forwards; 20 to 30 s: $\frac{1}{2}(10.0)(5.00) = 25$ m backwards.

Separation between the starting and final positions = 100 − 25 = 75.0 m

Explanation: After 20 s the velocity is negative, so the cyclist moves back towards the start. The question asks how far the final position is from the starting position, which is the displacement, 75.0 m. (The total path length travelled would be 100 + 25 = 125 m, option D.)

3. The coefficients of static friction and kinetic friction for a rubber block on flat wood are μ_s and μ_k respectively. What is the minimum horizontal force required for the rubber block of mass m starts to move from rest?

A $\mu_s mg$ B $\mu_k mg$ C $(\mu_s + \mu_k)\, mg$ D $(\mu_s - \mu_k)\, mg$

Answer: A

The block starts to move when the applied force just exceeds the maximum (limiting) static friction: $F = \mu_s R = \mu_s mg$.

Explanation: Before the block moves, static friction matches the applied force up to its maximum value $\mu_s mg$. Once it is moving, kinetic friction $\mu_k mg$ (smaller) acts, which is why it is easier to keep the block moving than to start it.

4. A ball moves from P and passes through Q on a frictionless surface as shown in the diagram below.



Which statement is true about the sum of gravitational potential and kinetic energies as the ball moves from P?

A The sum of energy decreases throughout the motion of the ball.
B The sum of energy remains unchanged throughout the motion of the ball.
C The sum of energy increases as the ball approaches Q and then decreases after passing Q.
D The sum of energy decreases as the ball approaches Q and then increases after passing Q.

Answer: B

On a frictionless surface no energy is lost, so the total mechanical energy (gravitational potential energy + kinetic energy) remains unchanged.

Explanation: As the ball goes down towards Q potential energy changes into kinetic energy, and as it climbs again kinetic energy changes back into potential energy, but their sum is constant (principle of conservation of mechanical energy).

5. The minutes hand of a wall clock is 8.2 cm and moves from 12 to 4. What is the angular displacement, θ, and the linear speed, v, of the minutes hand of the wall clock?

	θ/rad	v/cm s^{-1}
A	2.1	0.014
B	2.1	0.86
C	4.2	0.029
D	4.2	1.7

Answer: A

$$12 \text{ to } 4 \text{ is } \frac{4}{12} \text{ of a revolution: } \theta = \frac{1}{3}(2\pi) = 2.1 \text{ rad}$$

$$\omega = \frac{2\pi}{3600 \text{ s}} = 1.75 \times 10^{-3} \text{ rad s}^{-1}; \; v = r\omega = (8.2)(1.75 \times 10^{-3}) = 0.014 \text{ cm s}^{-1}$$

Explanation: The minute hand completes one revolution (2π rad) in 60 minutes = 3600 s. Keep the radius in cm so that the speed comes out in cm s^{-1}.

6. A particle moves in a horizontal circular motion, while another identical particle moves in a vertical circular motion with the same radius. Which are true about the motion of both particles?

	Horizontal circular motion	*Vertical circular motion*
A	The linear momentum is constant, but the kinetic energy varies	The linear momentum is constant, and the kinetic energy is constant
B	The linear momentum is constant, and the kinetic energy is constant	The linear momentum is constant, but the kinetic energy varies
C	The linear momentum varies, and the kinetic energy varies	The linear momentum varies, but the kinetic energy is constant
D	The linear momentum varies, but the kinetic energy is constant	The linear momentum varies, and the kinetic energy varies

Answer: D

Horizontal circle at constant speed: the speed (and kinetic energy) is constant but the direction of the velocity, and so the momentum, changes.

Vertical circle: gravity speeds the particle up on the way down and slows it on the way up, so both its momentum and its kinetic energy vary.

Explanation: Linear momentum is a vector, so it changes in any circular motion because the direction of motion changes. Kinetic energy is a scalar that depends only on speed: constant in uniform horizontal circular motion, but varying in a vertical circle where potential and kinetic energy interchange.

7. Two planets, A and B, have masses of M_A and M_B and radius of R_A and R_B respectively. Given that M_A is one quarter of M_B and the escape speed at the surface of A is one third the escape speed at the surface of B. What is the ratio of $\frac{R_A}{R_B}$?

A 0.19 B 0.44 C 2.25 D 5.33

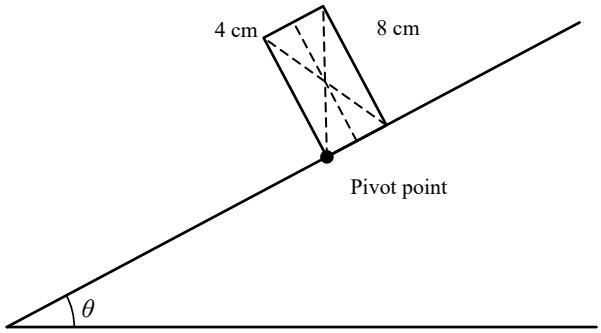
Answer: C

$$v = \sqrt{\left(\frac{2GM}{R}\right)} \Longrightarrow R = \frac{2GM}{v^2}$$

$$\frac{R_A}{R_B} = \frac{M_A}{M_B} \times \left(\frac{v_B}{v_A}\right)^2 = \frac{1}{4} \times 3^2 = 2.25$$

Explanation: Rearrange the escape speed formula for the radius, then form a ratio so that $2G$ cancels. The escape speed is squared, so the factor of 3 becomes 9.

8. A rectangular block of width 4 cm and length 8 cm is placed on a plane which inclined at an angle θ as shown in the diagram below.



Which statement is true about the block?

A The block will topple over if the angle θ is 25°.
B The block will not topple over because it has higher centre of gravity.
C The block will not topple over because its weight passes down through the pivot point.
D The block will topple over if the length of the block is placed on the plane which inclined at the same angle.

Answer: C

The block topples when the line of action of its weight falls outside its base, i.e. when $\tan\theta > \frac{\text{width}}{\text{height}} = \frac{4}{8} \Rightarrow \theta > 26.6°$.

In the diagram the weight acts through the pivot point, the limiting position, so the block does not topple.

Explanation: A block on a slope tips over the lower edge (the pivot). As long as the vertical line through the centre of gravity passes through or inside the base, the moment of the weight keeps the block on its base. At 25° (A) it is stable; with the 8 cm side on the slope (D) it would only topple above 63°.

9. Which statement is **not** one of the assumptions of the kinetic theory of an ideal gas?

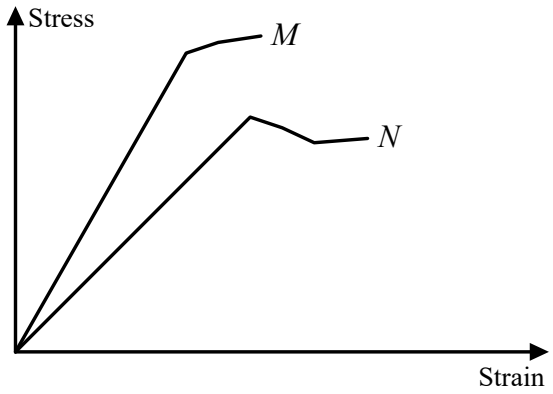
A Each collision is considered to be elastic.
B The volume of each molecule is negligible.
C All the molecules move with the same speed.
D There is no force between the molecules except during collision.

Answer: C

The molecules of an ideal gas move randomly with a range (distribution) of speeds; they do not all move with the same speed.

Explanation: The kinetic theory assumes: a large number of molecules in random motion, negligible molecular volume, perfectly elastic collisions, no intermolecular forces except during collisions, and collision times negligible compared with the time between collisions. The speeds follow the Maxwell–Boltzmann distribution.

10. The variation of stress with strain for materials, *M* and *N*, is shown in the graph below.



Which statement is true about the properties of *M* and *N*?

A *M* has greater value of Young modulus.
B *M* and *N* have an equal breaking point.
C *M* has greater proportional limit.
D *N* has greater tensile strength.

Answer: A

Young's modulus = gradient of the straight-line part of the stress–strain graph

The straight part of *M* is steeper than that of *N*, so *M* has the greater Young's modulus.

Explanation: A steeper initial gradient means a stiffer material: more stress is needed for the same strain. *M* also reaches a higher maximum stress, so *M* (not *N*) has the greater tensile strength, and the two materials break at different points.

11. The internal energy of a mol of an ideal gas of diatomic molecule at 500 K is 10.4 kJ. Which statement is true about the molecules?

A The molecules have rotational motion only.
B The molecules have five degrees of freedom.
C The molecules have three degrees of freedom.
D The molecules have translational motion only.

Answer: B

$$U = \frac{f}{2}nRT \Longrightarrow f = \frac{2U}{nRT} = \frac{2(10.4 \times 10^3)}{(1)(8.31)(500)} = 5$$

Five degrees of freedom: 3 translational + 2 rotational.

Explanation: For a diatomic molecule at ordinary temperatures the five degrees of freedom are three translational and two rotational, so the molecules have both translational and rotational motion (A and D are wrong).

12. What is the specific heat capacity of copper if the molar heat capacity is 24.5 J mol^{-1} K^{-1}? [Relative atomic mass of copper is 63.5.]

A 0.386 J kg^{-1} K^{-1} B 1.56 J kg^{-1} K^{-1} C 386 J kg^{-1} K^{-1} D 1560 J kg^{-1} K^{-1}

Answer: C

$$c = \frac{C_m}{M} = \frac{24.5 \text{ J mol}^{-1} \text{ K}^{-1}}{0.0635 \text{ kg mol}^{-1}} = 386 \text{ J kg}^{-1} \text{ K}^{-1}$$

Explanation: Divide the molar heat capacity by the molar mass in kg mol^{-1} (63.5 g mol^{-1} = 0.0635 kg mol^{-1}). Using 63.5 instead of 0.0635 gives option A.

13. A gas in a cylinder at constant pressure of 1.60×10^5 Pa is heated and the gas expands from 0.200 m^3 to 0.350 m^3. What is the work done by the gas?

A 2.4×10^4 J B 3.2×10^4 J C 5.6×10^4 J D 8.8×10^4 J

Answer: A

$$W = p\Delta V = (1.60 \times 10^5)(0.350 - 0.200) = 2.4 \times 10^4 \text{ J}$$

Explanation: At constant pressure the work done by the gas is the pressure multiplied by the change in volume (the area under the horizontal line on a *p*–*V* graph). Use the change in volume, 0.150 m^3, not the final volume.

14. The increase in emission of greenhouse gases is said to be the main cause of global warming. Which is the greenhouse gas?

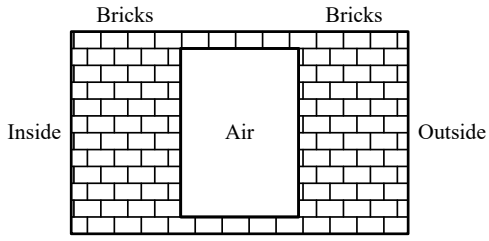
A Oxygen B Nitrogen C Hydrogen D Carbon dioxide

Answer: D

Carbon dioxide absorbs and re-emits infrared radiation from the Earth, so it is a greenhouse gas.

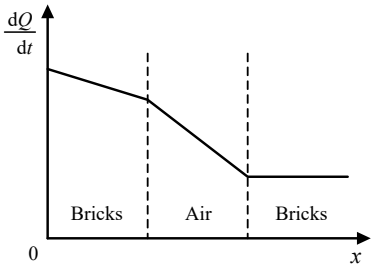
Explanation: Greenhouse gases (carbon dioxide, methane, water vapour, nitrous oxide) absorb infrared radiation. Oxygen, nitrogen and hydrogen are not greenhouse gases.

15. A section of a cavity wall made of bricks and air is shown in the diagram below.

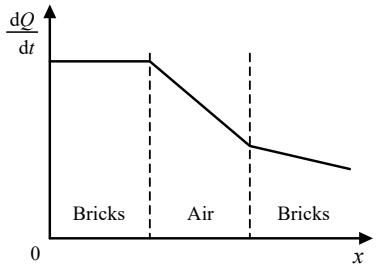


Each layer has the same thickness and the inside surface of the cavity wall is warmer than the outside surface. When a steady state is reached, which graph shows the variation of the rate of heat flow, $\frac{dQ}{dt}$, with the distance, x, from the inside surface of the wall?

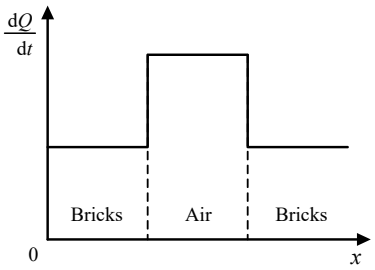
A



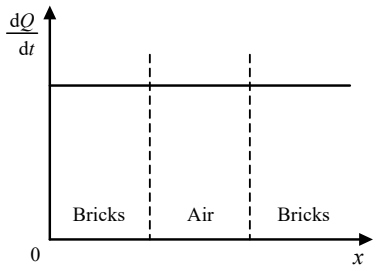
B



C



D



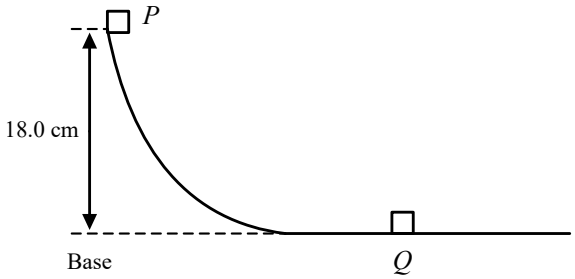
Answer: D

In the steady state the same rate of heat flow passes through every layer (bricks, air, bricks), because no heat is stored in any layer. So $\frac{dQ}{dt}$ is constant across the whole wall.

Explanation: Do not confuse the rate of heat flow with the temperature. The temperature falls through each layer (most steeply across the air gap, a poor conductor), but the rate of heat flow is the same everywhere in the steady state.

Section B [15 marks]

16. An object, *P*, of mass 0.025 kg is released from rest and slides down on a smooth curved track to the base and make a head-on collision with an identical object, *Q*, as shown in the diagram below.



P is released at height of 18.0 cm from the base and the velocity of Q after the collision is 1.41 m s^{-1}.

(a) Calculate the velocity of P before it collides with Q. [2]

Answer

$m_P gh = \frac{1}{2} m_P u_P^2 \Rightarrow u_P = \sqrt{(2gh)} = \sqrt{(2(9.81)(0.18))}$ 1

$u_P = 1.88$ m s^{-1} 1

[Accept 1.879 or 1.9 m s^{-1}]

2

Explanation: The track is smooth, so the potential energy lost while sliding down 18.0 cm (0.18 m) becomes kinetic energy. The mass cancels.

(b) Calculate the velocity of P after the collision. [2]

Answer

Conservation of momentum: $m_P u_P = m_P v_P + m_Q v_Q$; identical masses: $u_P = v_P + v_Q$ 1

$1.879 = v_P + 1.41 \Rightarrow v_P = 0.469$ m s^{-1} 1

2

Explanation: Q is initially at rest. With equal masses the masses cancel, and P keeps moving forwards with the remaining speed.

(c) Determine the type of the collision. [3]

Answer

Before: $E_i = \frac{1}{2}(0.025)(1.879)^2 = 0.0441$ J 1

After: $E_f = \frac{1}{2}(0.025)(1.41)^2 + \frac{1}{2}(0.025)(0.469)^2 = 0.0276$ J; $\Delta E = -0.0165$ J 1

Total kinetic energy is not conserved, so the collision is inelastic 1

3

Total for question **7**

Explanation: To decide the type of collision, compare the total kinetic energy before and after. Momentum is conserved in every collision, so it cannot be used to tell elastic from inelastic collisions.

17. (a) Describe heat transfer by radiation. [2]

Answer

• Radiation is heat transfer by electromagnetic waves (infrared) 1

• The energy travels at the speed of light 1

• No medium is required 1

[Any two – maximum 2 marks]

2

Explanation: Radiation is the only method of heat transfer that works through a vacuum, which is how the Sun's energy reaches the Earth.

(b) A tungsten sphere of radius 15.0 cm and emissivity 0.35 at 20.0 °C is transferred into a cavity with a maintained surrounding temperature at −4.0 °C.

(i) Determine the power radiated by the sphere. [3]

Answer

$P = \sigma A e T^4$ 1

$A = 4\pi r^2 = 4\pi(0.150)^2$; $P = (5.67 \times 10^{-8})(4\pi)(0.150)^2(0.35)(293)^4$ 1

$P = 41.4$ W 1

3

Explanation: Use the surface area of the sphere and the temperature of the sphere in kelvin (20.0 + 273 = 293 K). The emissivity 0.35 shows tungsten is not a black body.

(ii) Determine the net rate of energy transfer of the sphere. [2]

Answer

$\frac{dQ}{dt} = P_{out} - P_{in} = \sigma A e(T_2^4 - T_1^4) = (5.67 \times 10^{-8})(4\pi)(0.150)^2(0.35)(293^4 - 269^4)$ 1

Net rate = 12.0 J s^{-1} (W) 1

2

Explanation: The sphere emits radiation and also absorbs radiation from the surroundings at −4.0 °C (269 K). The net rate of energy loss is the difference. Raise each temperature to the fourth power before subtracting.

(iii) What will happen to the power radiated by the sphere when the radius of the tungsten sphere is increased? [1]

Answer

The power radiated by the sphere increases	1
	1
Total for question	**8**

Explanation: The power radiated is proportional to the surface area, which is proportional to r^2; a larger sphere at the same temperature radiates more.

Section C [30 marks]

18. (a) State Newton's law of universal gravitation. [2]

Answer

Every particle of matter in the universe attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.	2
	2

Explanation: Both proportionalities and the word attracts are needed. In symbols: $F = \frac{Gm_1m_2}{r^2}$.

(b) Explain the relation between high tides phenomenon and the position of the Earth which aligns between the Moon and the Sun. [4]

Answer

High tides are caused mainly by the gravitational force of the Moon on the Earth (and its oceans)	1
The gravitational force is inversely proportional to the square of the distance: $F \propto \frac{1}{r^2}$	1
When the Earth is aligned between the Moon and the Sun (full moon), the Moon and the Sun pull along the same line and the Moon is closest to the Earth (r_{EM} smallest)	1
So the resultant gravitational pull on the Earth is the largest, giving the highest tides	1
	4

Explanation: Tides are produced by the difference in gravitational pull across the Earth. When the Sun, Earth and Moon are in a straight line their tidal effects add, producing very high tides (spring tides).

(c) A satellite is orbiting the Earth in a circular path with a duration of 6.2×10^3 s. The mass of the satellite is 5.5×10^3 kg. Determine

(i) the radius of the orbit, [5]

Answer

$F = \frac{mv^2}{r}$ 1

$v = r\omega$ 1

$\omega = \frac{2\pi}{T} \Longrightarrow F = \frac{4\pi^2 mr}{T^2}$ 1

Gravitational force provides it: $\frac{GM_e m}{r^2} = \frac{4\pi^2 mr}{T^2} \Longrightarrow r = \left(\frac{GM_e T^2}{4\pi^2}\right)^{1/3}$ 1

$r = \left(\frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(6.2 \times 10^3)^2}{4\pi^2}\right)^{1/3} = 7.29 \times 10^6$ m 1

5

Explanation: The only force on the satellite is the Earth's gravity, which provides the centripetal force. Writing the speed in terms of the period gives Kepler's third law, $T^2 \propto r^3$. The mass of the satellite cancels.

(ii) the weight of the satellite, [2]

Answer

$W = F_g = \frac{GMm}{r^2} = \frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(5.5 \times 10^3)}{(7.292 \times 10^6)^2}$ 1

$W = 4.12 \times 10^4$ N 1

2

Explanation: The weight of the satellite in orbit is the gravitational force on it at that radius, not its weight on the Earth's surface (5.4×10^4 N). Astronauts feel weightless because they are in free fall, not because there is no gravity.

(iii) the potential energy of the satellite. [2]

Answer

$U = -\frac{GMm}{r} = -\frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(5.5 \times 10^3)}{7.292 \times 10^6}$ 1

$U = -3.00 \times 10^{11}$ J 1

2

Total for question **15**

Explanation: Gravitational potential energy is negative, with zero at infinity. Note that it is divided by r, not r^2.

19. (a) State **two** differences between static friction and kinetic friction. [2]

Answer

• Static friction acts on a stationary object; kinetic friction acts on an object sliding on a surface	1
• Static friction varies with the applied force (up to a maximum); kinetic friction is constant	1
• The limiting (maximum) static friction is greater than kinetic friction	1
[Any two – maximum 2 marks]	
	2

Explanation: Static friction adjusts itself to stop motion starting, up to the limiting value $\mu_s R$. Once sliding starts, kinetic friction has a fixed, smaller value $\mu_k R$.

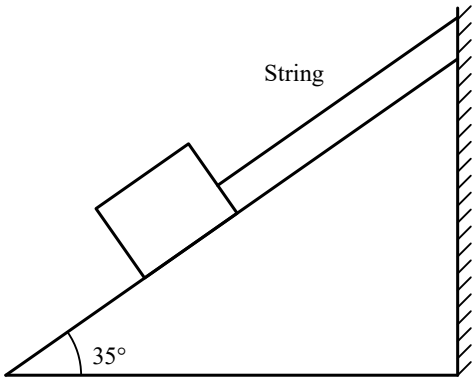
(b) A box of mass 10.0 kg rests on an inclined plane. The inclination angle of the plane, θ, is increased slowly until the box begins to slide. The coefficient of static friction between the box and the inclined plane is 0.42. Determine θ at the instant the box is just about to slide. [4]

Answer

Just about to slide: $mg \sin\theta = f_s$	1
$f_s = \mu_s mg \cos\theta$	1
$\tan\theta = \mu_s$	1
$\theta = \tan^{-1} 0.42 = 22.8°$	1
	4

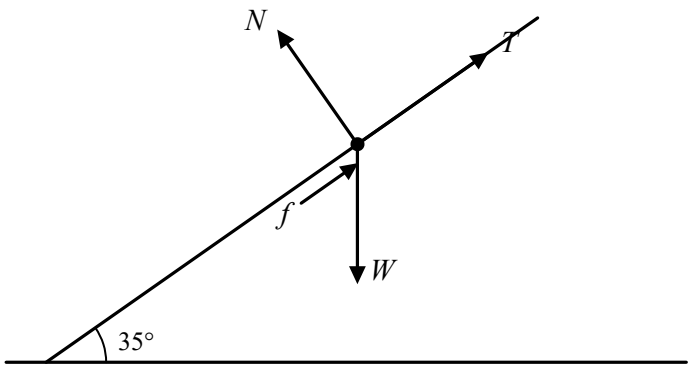
Explanation: At the angle of repose the component of the weight down the slope just equals the maximum static friction. The mass and g cancel, so the angle depends only on the coefficient of static friction.

(c) The inclination angle of the plane in (b) is raised to 35° with the horizontal and a string parallel to the inclined plane is used to keep the box in equilibrium as shown in the diagram below.



(i) Sketch a free body diagram and label all the forces acting on the box. [3]

Answer



Normal reaction N perpendicular to the plane and weight W vertically downwards	1
Tension T up the plane, parallel to the string	1
Friction f up the plane	1
	3

Explanation: At 35° (more than 22.8°) the box would slide down without the string, so friction acts up the slope. For the minimum tension, friction is at its maximum value and helps the string to hold the box.

(ii) Calculate the minimum tension in the string. [2]

Answer

$T_{\min} + f_s = mg \sin 35°; f_s = \mu_s mg \cos 35° = (0.42)(10.0)(9.81) \cos 35° = 33.75$ N	1
$T_{\min} = (10.0)(9.81) \sin 35° - 33.75 = 22.5$ N	1
	2

Explanation: The tension is smallest when static friction is at its limiting value. Together they balance the component of the weight down the slope, 56.3 N.

(iii) If the string is cut, determine the acceleration of the box which slides down the plane. Given that the coefficient of kinetic friction is 0.35. [4]

Answer

$f_k = \mu_k mg \cos 35° = (0.35)(10.0)(9.81) \cos 35° = 28.13$ N	1
Resultant force down the plane: $F = mg \sin 35° - f_k$	1
$F = ma$: $(10.0)a = 56.27 - 28.13$	1
$a = 2.81$ m s^{-2}	1
	4
Total for question	**15**

Explanation: When the string is cut the box slides, so kinetic friction (not static) acts up the slope. Newton's second law gives the acceleration down the plane.

20. (a) Describe the thermodynamic changes in isothermal expansion and adiabatic expansion. [7]

Answer

$Q = \Delta U + W$	1
Isothermal process: the temperature is constant, $\Delta T = 0$	1
$\Delta U = 0$, so $Q = W$	1
In the expansion, the heat supplied is used to do work	1
Adiabatic process: no heat enters or leaves the system	1
$Q = 0$, so $\Delta U = -W$	1
In the expansion, the internal energy decreases (the gas cools)	1
	7

Explanation: Apply the first law to each process. The internal energy of an ideal gas depends only on temperature: it is constant in an isothermal process. In an adiabatic expansion the work done by the gas comes from its internal energy.

(b) A cylinder with adjustable piston contains 3.50 mol of diatomic gas is initially at 0.125 m^3 and 1.013×10^5 Pa. The volume of the gas is then increased adiabatically to 0.85 m^3. Determine the

(i) initial and final temperatures of the gas, [5]

Answer

$PV = nRT \Longrightarrow T_1 = \frac{PV}{nR} = \frac{(1.013 \times 10^5)(0.125)}{(3.50)(8.31)}$ 1

$T_1 = 435.4$ K 1

$TV^{\gamma-1} = \text{constant}$ 1

$\gamma = 1.4$ (diatomic): $T_2 = T_1(\frac{V_1}{V_2})^{\gamma-1} = 435.4(\frac{0.125}{0.85})^{0.4}$ 1

$T_2 = 202.2$ K 1

5

Explanation: The initial temperature comes from the ideal gas equation. For the adiabatic expansion use $TV^{\gamma-1} = \text{constant}$ with $\gamma = \frac{7}{5}$ for a diatomic gas; the gas cools because it does work without receiving heat.

(ii) change in internal energy of the gas. [3]

Answer

$\Delta U = \frac{f}{2} nR\Delta T$ 1

$f = 5$: $\Delta U = \frac{5}{2}(3.50)(8.31)(202.2 - 435.4)$ 1

$\Delta U = -1.70 \times 10^4$ J 1

3

Total for question **15**

Explanation: The change in internal energy depends only on the change in temperature. The negative sign shows that the internal energy decreases, which is equal to the work done by the gas in the adiabatic expansion.